Population Dynamics and Projections - The Cohort Component Method

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1. Key Definitions

Estimation: calculation of any demographic measurement given a set of assumptions (e.g., rates).
Projection: estimation of the size and distribution of the population at one time from the size and distribution at another, together with a specification of how change takes place in between.
Forecast: a projection based on predictive assumptions of future population trends.
Scenario: a projection based on explicit assumptions about some specific population trends.

2. Motivation for Projections

Short-term projections are good for budget plans (local and national governments).
Long-term projections have several applications: urban and city planning; economic valuation; sustainability of natural resources; some pension schemes.
Long-term projections are good for anticipating future demographic trends (ageing); and evaluate policy interventions such as contraceptive programs.
Population projections are carried out by demographers at national statistical offices or international agencies.
Demographers in academic positions evaluate the accuracy of these projections and develop new methods to increase the flexibility and precision of population projections.

3. Mathematical Methods for Projections (the alternative view to the cohort component method)

Geometric growth - annual compounding:

$$P_t = P_0 \cdot (1 + r)^t. \tag{1}$$

Geometric growth - continuous compounding:

$$P_t = P_0 \cdot e^{r \cdot t}. \tag{2}$$

The logistic curve:

$$P_t = \frac{P_\infty}{1 + e^{-s \cdot (t-h)}}. \tag{3}$$

P_∞ is the asymptotic value of the population (*ad infinitum*); h is the time when the population is half the ultimate size; and s is the maximum speed of growth.

Polynomial growth model:

$$P_t = P_0 + a_1 \cdot t + a_2 \cdot t^2 + ... + n \cdot t^n. \tag{4}$$

Advantages:

Mathematical methods are easy to apply, with minimal data requirements (few observations).

Disadvantages:

Mathematical methods are crude extrapolations ignoring demographic components, population groups, subgroups, and logical interdependence. For example, the number of births (i.e., the size of a new cohort) depends on age-specific fertility rates and the size and age distribution of the female population in reproductive ages.

Appropriate methods:

When vital rates and age structure can reasonably be assumed to remain constant—for example, in the very short term or in the case of a stable population.
When only the total population size is known.

4. The Cohort Component Method (Overview)

Accounts for population dynamics, not just the overall population growth—e.g. age structure and sex composition.
Ideal when these are important features for planning and policy.
Accounts for fact that as populations grow, their age structure is likely to change and this will affect the birth rate and death rate (producing fluctuations in growth).
Developed by Pascal K Whelpton in the 1930s.

Advantages:

Models not only total population size but also the dynamics of the population structure (age–sex composition).
Makes explicit the three components of demographic change: fertility, mortality, and migration (in open populations).
Allows segmentation of the population into sub-groups with different fertility, mortality, and migration rates, and enables separate computation of changes over time for each group.
Conceptually simple, but can become complex when incorporating multiple sub-groups or changing assumptions about demographic components.

Balancing equation:

Only two possible ways of joining a population: birth or in-migration. The only ways to leave a population are to emigrate or to die.

$$N(t+n) = N(t) + B[t, t+n] - D[t, t+n] + I[t, t+n] - O[t, t+n]. \tag{5}$$

Cohort:

Individuals of the same generation (born in the same year) will experience the same demographic events at the same time. They are affected by the same rates of mortality, fertility, and migration at the same time; even if these rates change by age and time.

Components:

A population model depending on the three sources of demographic change (age-specific rates of mortality, fertility, and migration).

Age changes at same speed as time:

One year of time is one year of age. The population of age x, at time t, is the surviving population of age $x-1$, at time $t-1$, plus the effect of the net migrants. **Projection intervals should be the same length as the age intervals** (usually 5 years, or single years).

Data requirements:

Base population by age and sex. Population will be projected at the midpoint of each time interval.
Life tables for each sex. One life table for each projection interval. Predicted mortality rates depending on the forecast of a mortality index. For example, the expected value of the life expectancy at birth during the following years.
Age-specific fertility rates for each projection interval. Necessary information to calculate these rates can be a good substitute. For example, estimated number of births by groups of age during a period of time or a forecast TFR.
Age- and sex-specific (net) migration rates for each projection interval. Necessary information to calculate these rates can be a good substitute. For example, the estimated number of net migrants during a period of time.
An assessment of the **Sex Ratio at Birth** (as a substitute for maternity rates).
Models can be used as a substitute, when fertility and mortality rates are unreliable or incomplete.

5. The Cohort Component Method (Formalisation)

Different equations are used for the first age interval, the open-ended age interval, and all intermediate age intervals. Solutions are presented from the simplest to the most complex case. Let's assume a **female-dominant perspective**: women reproduce the population, and the number of births is proportional to the number and age distribution of women in the reproductive ages. Let's focus on the projected number of women, keeping in mind that women give birth to both boys and girls. **The male population is projected using different inputs but the same equations**. For now, let's assume the population is closed to migration—an assumption that will be relaxed in this lecture.

All intermediate age intervals (closed population)

The (female) population of age x at the time $t+n$ is equal to the survivors who had $x-n$ years at the time t, calculated by multiplying the population at the time t by the corresponding survivorship ratio, as shown:

$$ {}_nN_x^f(t+n) = {}_nN_{x-n}^f(t) \cdot \frac{{}_nL_x^f}{{}_nL_{x-n}^f} \tag{6} $$

In this case x is greater than zero and less than the open-ended age interval.

The open-ended age interval (closed population)

The projected open-ended age interval consists of two populations: those who were already in the open-ended age interval at time t, and those who were not in it at t but will enter it by $t+n$.

$$ {}_\infty N_x^f(t+n) = \left[{}_\infty N_x^f(t) + {}_nN_{x-n}^f(t)\right] \cdot \frac{{}_\infty L_x^f}{{}_\infty L_x^f + {}_nL_{x-n}^f} \tag{7} $$

In this case x denotes the beginning of the open ended age interval.

The first age interval (closed population)

The population of the first age interval is projected in two steps. First, we calculate the projected number of births during the projection interval using age-specific fertility rates and the age distribution of the population

in the reproductive ages. To distinguish between male and female births, we can use either maternity rates for female births or the **Sex Ratio at Birth**. The projected number of births is then multiplied by the corresponding survivorship ratio for boys and girls.

Key Assumption: The projected number of births in the age interval $[x, x+n]$, is a function of the age-specific fertility rate ${}_nF_x$, and the number of person-years lived at those ages during the time interval $[t, t+n]$, i.e., $events = rate \cdot exposure$. The latter is calculated as the average number of women aged x to $x+n$ who are alive at the beginning and at the end of the interval (i.e., assuming linear growth and a mid-period approximation). The total number of births is then calculated as follows, by adign the number of birth at all ages:

$$B[t, t+n] = \sum_{x=\alpha}^{\beta-n} n \cdot {}_nF_x \cdot \frac{1}{2} \cdot \left[{}_nN_x^f(t) + {}_nN_{x-n}^f(t) \cdot \frac{{}_nL_x^f}{{}_nL_{x-n}^f} \right] \tag{8}$$

The Sex Ratio at Birth is used to calculate the projected number of male and female births:

$$B^f[t, t+n] = \frac{1}{1+SRB} \cdot B[t, t+n] \tag{9}$$

The projected population in the first age interval is then calculated from the projected number of births and the corresponding survivorship ratio, whose denominator is n times the radix of the life table:

$${}_nN_0^f(t+n) = B^f[t, t+n] \cdot \frac{{}_nL_0^f}{n \cdot l_0^f} \tag{10}$$

Male population is projected using similar equations. However, in a **female-dominant projection**, the number of births depends on the size, age distribution, and the fertility rates of the female population. In the simplest form, projected male births are obtained by multiplying the number of female births by the SRB.

Migration

Under some conditions, the most conservative assumption is to project the population ignoring the effect of migration. Migration could be highly unpredictable and perhaps, the most complex of the three sources of population change. However, migrations are increasing and have a significant impact on the size and distribution of populations; for example, in contexts of low fertility and low mortality.

The system of equations 6, 7, 8, 9, and 10—to project a closed population—can be modified to account for the effect of net migration. **There are different approaches to model migration**. One leading approach is to assume that one half immigrate/emigrate at the beginning and the second half at the end of the time interval. While the first portion is affected by the fertility and mortality schedules, the second portion would be independent. This is not a restrictive assumption when the value of n is small.

All intermediate age intervals (open population)

The (female) population of age x at the time $t+n$ is equal to the survivors who had $x-n$ years at the time t plus the half of net migrants arriving at time $t+n$. However, the population at the time t is already augmented by one half the number of net migrants I, as shown:

$${}_nN_x^f(t+n) = \left[{}_nN_{x-n}^f(t) + \frac{1}{2} \cdot {}_nI_{x-n}^f[t, t+n] \right] \cdot \frac{{}_nL_x^f}{{}_nL_{x-n}^f} + \frac{1}{2} \cdot {}_nI_x^f[t, t+n] \tag{11}$$

Note that those net migrants arriving at time t are n years younger than those arriving at time $t+n$. The implicit assumption is that net migrants will follow the mortality schedule of the receiving population. This assumption can be relaxed by using a different life table for net migrants. For example, if net migrants are more (or less) healthier than the receiving population.

The open-ended age interval (open population)

Following the same logic as for the intermediate age intervals, the two populations in the open-ended age interval are increased (or reduced) by the number of net migrants arriving (or departing) at exact time t

and surviving under the receiving mortality schedule, as well as by the number of net migrants arriving (or departing) at exact time $t+n$ who are already in the open-ended age interval, as shown:

$$_{\infty}N_x^f(t+n) = \left[{}_{\infty}N_x^f(t) + {}_nN_{x-n}^f(t) + \frac{1}{2} \cdot \left[{}_nI_{x-n}^f[t,t+n] + {}_{\infty}I_x^f[t,t+n] \right] \right] \cdot \frac{{}_{\infty}L_x^f}{{}_{\infty}L_x^f + {}_nL_{x-n}^f} + \frac{1}{2} \cdot {}_{\infty}I_x^f[t,t+n] \tag{12}$$

The first age interval (open population)

In an open population, migrant women of reproductive age will contribute to the projected number of births. Under the same assumptions applied to calculate exposure to childbearing (growth is linear and using a mid-period approximation), the projected number of births is computed as follows:

$$B[t,t+n] = \sum_{x=\alpha}^{\beta-n} n \cdot {}_nF_x \cdot \frac{1}{2} \cdot \left[{}_nN_x^f(t) + \frac{1}{2} \cdot {}_nI_x^f[t,t+n] + {}_nN_x^f(t+n) \right] \tag{13}$$

$$B[t,t+n] = \sum_{x=\alpha}^{\beta-n} {}_nF_x \cdot \frac{n}{2} \cdot \left[{}_nN_x^f(t) + {}_nI_x^f[t,t+n] + \left[{}_nN_{x-n}^f(t) + \frac{1}{2} \cdot {}_nI_{x-n}^f[t,t+n] \right] \cdot \frac{{}_nL_x^f}{{}_nL_{x-n}^f} \right] \tag{14}$$

Note that if a population has a negative net migration, the number of births will be reduced. In some contexts, assuming that migrant women would follow the same fertility schedule of the receiving population could be restrictive or unrealistic. This equation can be adjusted to account for different fertility schedules when migrants are fertility selected and this information is available, or simply to quantify the effect of migration on childbearing.

$$B[t,t+n] = \sum_{x=\alpha}^{\beta-n} {}_nF_x \cdot \frac{n}{2} \cdot \left[{}_nN_x^f(t) + {}_nN_{x-n}^f(t) \cdot \frac{{}_nL_x^f}{{}_nL_{x-n}^f} \right] + \sum_{x=\alpha}^{\beta-n} {}_nF_x \cdot \frac{n}{2} \cdot \left[{}_nI_x^f[t,t+n] + \frac{1}{2} \cdot {}_nI_{x-n}^f[t,t+n] \cdot \frac{{}_nL_x^f}{{}_nL_{x-n}^f} \right] \tag{15}$$

As in the case of a closed population, the SRB is used to calculate the projected number of male and female births:

$$B^f[t,t+n] = \frac{1}{1+SRB} \cdot B[t,t+n] \tag{16}$$

Finally, the projected population in the first age interval is calculated from the projected number of births and the corresponding survivorship ratio, increased by one-half of the number of migrants who arrive at time $t+n$ in that same age interval, as shown:

$$_nN_0^f(t+n) = B^f[t,t+n] \cdot \frac{{}_nL_0^f}{n \cdot l_0^f} + \frac{1}{2} \cdot {}_nI_0^f[t,t+n] \tag{17}$$

Male population is projected using similar equations and the same underlying assumptions: the number of births depends on the size and distribution of the female population at reproductive age; one half of migrants would arrive at the beginning and the other half at the end of the projection interval.

Summary:

Cohort-component models require—and make full use of—information on the components of population change (fertility, mortality, and migration).
The model provides estimates of the future population by age and sex.
Calculations are more complex than mathematical extrapolation of the total population.
The projection period is divided into intervals that match the length of the age intervals.
All age groups should have the same width, except for the open-ended last age group.
Projections proceed one interval at a time.
The procedure involves multiple steps: survivorship, reproduction, and the addition of net migrants.

6. The Cohort Component Method (The Leslie Matrix Approach)

Advantages of matrix notation:

Compact and elegant representation of the whole system of equations.
Efficient computation that facilitates the implementation and scalability in a programming language (e.g., several populations, several rounds of projection, and several scenarios).

The Leslie matrix:

The Leslie matrix is a square matrix whose order equals the number of age groups. The first row contains the age-specific fertility rates, and the sub-diagonal contains the survivorship ratios.
Let's assume a population consisting of four age groups of equal length plus a final open-ended age interval. Let's also assume that the population includes only one sex, reproduces in age groups 2, 3, and 4, and is closed to migration. The Leslie matrix $\mathbf{L}$ for this population would be as follows:

$$\mathbf{L} = \begin{bmatrix} 0 & \frac{{}_nL_0}{n \cdot l_0} \cdot \frac{n}{2} \cdot \left[{}_nF_n + {}_nF_{2n} \cdot \frac{{}_nL_{2n}}{{}_nL_n} \right] & \frac{{}_nL_0}{n \cdot l_0} \cdot \frac{n}{2} \cdot \left[{}_nF_{2n} + {}_nF_{3n} \cdot \frac{{}_nL_{3n}}{{}_nL_{2n}} \right] & \frac{{}_nL_0}{n \cdot l_0} \cdot \frac{n}{2} \cdot {}_nF_{3n} & 0 \\ \frac{{}_nL_n}{{}_nL_0} & 0 & 0 & 0 & 0 \\ 0 & \frac{{}_nL_{2n}}{{}_nL_n} & 0 & 0 & 0 \\ 0 & 0 & \frac{{}_nL_{3n}}{{}_nL_{2n}} & 0 & 0 \\ 0 & 0 & 0 & \frac{{}_\infty L_{4n}}{{}_\infty L_{4n} + {}_nL_{3n}} & \frac{{}_\infty L_{4n}}{{}_\infty L_{4n} + {}_nL_{3n}} \end{bmatrix} \tag{18}$$

To relax the notation, we can use the function ${}_nS_x$, to denote the survivorship ratio:

$${}_nS_{x+n} = \frac{{}_nL_{x+n}}{{}_nL_x}, \tag{19}$$

and the function ${}_nG_x$, to indicate how women of the same cohort would transit from one age-specif fertility rate to another as their age increase from x to $x+n$, to the form:

$${}_nG_x = \frac{n}{2} \left[{}_nF_x + {}_nF_{x+n} \cdot \frac{{}_nL_{x+n}}{{}_nL_x} \right]. \tag{20}$$

The resulting Leslie matrix, presented in compact notation, is described as follows:

$$\mathbf{L} = \begin{bmatrix} 0 & {}_nS_0 \cdot {}_nG_n & {}_nS_0 \cdot {}_nG_{2n} & {}_nS_0 \cdot {}_nG_{3n} & 0 \\ {}_nS_n & 0 & 0 & 0 & 0 \\ 0 & {}_nS_{2n} & 0 & 0 & 0 \\ 0 & 0 & {}_nS_{3n} & 0 & 0 \\ 0 & 0 & 0 & {}_\infty S_{4n} & {}_\infty S_{4n} \end{bmatrix} \tag{21}$$

Hence, the projected population at time $t+n$ is equal to the product of the Leslie matrix and the population at time t:

$$\mathbf{N}(t+n) = \mathbf{L}\,\mathbf{N}(t) \tag{22}$$

$$\begin{bmatrix} {}_nN_0(t+n) \\ {}_nN_n(t+n) \\ {}_nN_{2n}(t+n) \\ {}_nN_{3n}(t+n) \\ {}_\infty N_{4n}(t+n) \end{bmatrix} = \begin{bmatrix} 0 & {}_nS_0 \cdot {}_nG_n & {}_nS_0 \cdot {}_nG_{2n} & {}_nS_0 \cdot {}_nG_{3n} & 0 \\ {}_nS_n & 0 & 0 & 0 & 0 \\ 0 & {}_nS_{2n} & 0 & 0 & 0 \\ 0 & 0 & {}_nS_{3n} & 0 & 0 \\ 0 & 0 & 0 & {}_\infty S_{4n} & {}_\infty S_{4n} \end{bmatrix} \begin{bmatrix} {}_nN_0(t) \\ {}_nN_n(t) \\ {}_nN_{2n}(t) \\ {}_nN_{3n}(t) \\ {}_\infty N_{4n}(t) \end{bmatrix} \tag{23}$$

If a population is projected for 5 consecutive periods of n years each, the population at time $t+5n$ is calculated as follows:

$$\mathbf{N}(t+5n) = \mathbf{L}^5\,\mathbf{N}(t) \tag{24}$$

A Leslie matrix for an open population:

Keeping the same assumptions previously discussed regarding fertility and migration, the first adjustment to consider is the effect of migration on the projected number of births and on reproduction in general. The first step is to calculate age-specific net migration rates by dividing the number of net migrants by the population at time t. This is not a restrictive assumption, given that half of all migrants are assumed to arrive (or depart) at the at the exact time t, as shown:

$$ {}_nK_x = \frac{{}_nI_x[t, t+n]}{{}_nN_x(t)}. \tag{25} $$

The second step is to quantify the effect of migration on the projected number of births through fertility and reproduction, which requires modifying the function ${}_nG_x$.

$$ {}_nG_x = \frac{n}{2}\left[(1 + {}_nK_x) \cdot {}_nF_x + \left(1 + \frac{1}{2} \cdot {}_nK_x\right) \cdot {}_nF_{x+n} \cdot \frac{{}_nL_{x+n}}{{}_nL_x}\right]. \tag{26} $$

The third step is to modify the first sub-diagonal of the Leslie matrix using the survivorship ratios but, now, accounting for the effect of migration; and the main diagonal, which now quantifies the effect of migrants who arrive at the end of the projection period, as shown:

$$ \mathbf{L} = \begin{bmatrix} \frac{{}_nK_0}{2} & {}_nS_0 \cdot {}_nG_n & {}_nS_0 \cdot {}_nG_{2n} & {}_nS_0 \cdot {}_nG_{3n} & 0 \\ \left(1 + \frac{{}_nK_n}{2}\right) \cdot {}_nS_n & \frac{{}_nK_n}{2} & 0 & 0 & 0 \\ 0 & \left(1 + \frac{{}_nK_{2n}}{2}\right) \cdot {}_nS_{2n} & \frac{{}_nK_{2n}}{2} & 0 & 0 \\ 0 & 0 & \left(1 + \frac{{}_nK_{3n}}{2}\right) \cdot {}_nS_{3n} & \frac{{}_nK_{3n}}{2} & 0 \\ 0 & 0 & 0 & \left(1 + \frac{{}_\infty K_{4n}}{2}\right) \cdot {}_\infty S_{4n} & \left(1 + \frac{{}_\infty K_{4n}}{2}\right) \cdot {}_\infty S_{4n} + \frac{{}_\infty K_{4n}}{2} \end{bmatrix} \tag{27} $$

Once the Leslie matrix accounts for migration, the projected population at time $t + n$ is equal to the product of the Leslie matrix and the population at time t:

$$ \mathbf{N}(t+n) = \mathbf{L}\,\mathbf{N}(t) \tag{28} $$

Exercise 1: Cohort Component Model of Projections (one sex, female dominant perspective, closed population)

The heading of the R script

The R script begins by clearing the workspace. It requires the **stringr** package for string manipulation (e.g., symbols, letters, and numbers), **Matrix** for matrix operations, and **expm** for exponential matrix calculation—please ensure these packages are installed.

```
rm(list = ls())
#install.packages("stringr")
#install.packages("Matrix")
#install.packages("expm")
library(stringr)
library(Matrix)
library(expm)
```

The data:

In this exercise, we are given population counts ${}_nN_x$, age-specific fertility rates ${}_nF_x$, and the number of person-years lived in each age interval ${}_nL_x$ representing the mortality conditions of the female population of Sweden in 1993 (January 1st), using 5-year age intervals, as reported by Preston et al. 2001. "Population projections." Pp. 117-137 in *Demography: measuring and modeling population processes*. Malden, MA: Blackwell Publishers. Data are arranged as a matrix, and each column represents one of the three essential

inputs for the analysis. For ease of access, data have been temporarily stored in a GitHub repository and can be retrieved using the following lines of code.

```
GitHub            = "https://raw.githubusercontent.com/Romero-Prieto/teaching/main/Population%20Dynamics
input             = read.csv(GitHub)
```

To simplify data preparation, we can read the data as a matrix and create a table that includes only the relevant information.

```
input             = as.matrix(input)
female            = input[,c("age","nLx_f","nFx","nNx_f")]
```

Definitions:

It is useful to define some parameters before beginning the calculations. These parameters are not strictly necessary, but they make the code more readable and easier to modify if needed. Those parameters are: the baseline year, the number of age groups, the radix of the life table, the length of the age intervals—which is the same for all—and the sex ratio at birth.

```
year              = 1993.0
ages              = nrow(female)
radix             = 100000
n                 = 5
SRB               = 1.05
```

Construction of the Leslie matrix:

The first sub-diagonal of the Leslie matrix contains the survivorship ratios, which are calculated from the ${}_nL_x$ column of a life table. We can generate a variable ${}_nL_x$ and calculate the survivorship ratios ${}_nS_x$, keeping in mind that different formulas are used for the first age interval, the open-ended age interval, and all intermediate age intervals.

```
nLx               = female[,"nLx_f"]
nSx               = c(nLx[1]/(n*radix),nLx[2:(ages-1)]/nLx[1:(ages-2)],nLx[ages]/sum(nLx[(ages-1):ages]
```

To calculate the projected number of births, the cohort component model makes specific assumptions about the fertility schedule of the population. We can define a function ${}_nG_x$ to incorporate these assumptions.

```
nFx               = female[,"nFx"]
nGx               = (nFx + c(nFx[2:ages]*nSx[2:ages],0))*n/2
```

We define the Leslie matrix as a square matrix whose order is equal to the number of age groups. Note that, before the matrix is populated, all entries are set to zero. The first row contains a set of multipliers related to reproduction, determined by the survivorship ratio of the first age interval, the fertility schedule, and the sex ratio at birth, as shown:

```
lESliE            = matrix(0,ages,ages)
lESliE[1,]        = nGx*1/(1 + SRB)*nSx[1]
```

We then populate the sub-diagonal with the survivorship ratios for all intermediate age intervals, and place the survivorship ratio for the open-ended age interval in the last entry of the last column, as shown:

```
diag(lESliE[-1,]) = nSx[2:ages]
lESliE[ages,ages] = nSx[ages]
```

Projecting the female population:

Matrix multiplication simplifies the projection to a single operation. We simply multiply the Leslie matrix by the population vector at time t to obtain the projected population at time $t + n$. Note that R uses a very specif notation for matrix multiplication (i.e., the operator %*%), which is different from the notation for the element-wise multiplication (i.e., the operator *).

```
current           = female[,"nNx_f"]
projected         = lESliE%*%current
```

Let's produce a simple table of results by projecting the population for three consecutive periods while holding fertility and mortality constant (i.e., using the same Leslie matrix throughout). We first retain only the relevant information and rename the current population using the reference year.

```
female            = input[,c("age","nNx_f")]
colnames(female)[colnames(female) == "nNx_f"] <- str_c("nNx_",sprintf("%.1f", year))
```

We then use a loop to project the population iteratively, given the most recent data at each step, as shown:

```
for (t in seq(year, year + 2*n, n)) {
  projected = lESliE%*%female[,str_c("nNx_",sprintf("%.1f", t))] #Projects
  colnames(projected)[1] <- str_c("nNx_",sprintf("%.1f", t + n)) #Gives a name to the projected data.
  female = cbind(female,projected)                               #Concatenates the table.
}
```

Finally, we print the resulting table:

```
print(female)
```

```
##       age nNx_1993.0 nNx_1998.0 nNx_2003.0 nNx_2008.0
##  [1,]   0     293395   293573.8   280120.5   264190.5
##  [2,]   5     248369   293189.2   293367.9   279924.0
##  [3,]  10     240012   248250.6   293049.4   293228.0
##  [4,]  15     261346   239833.3   248065.7   292831.2
##  [5,]  20     285209   261014.9   239529.5   247751.5
##  [6,]  25     314388   284786.9   260628.6   239174.9
##  [7,]  30     281290   313781.7   284237.6   260125.9
##  [8,]  35     286923   280463.0   312859.1   283401.9
##  [9,]  40     304108   285576.2   279146.5   311390.6
## [10,]  45     324946   301730.7   283343.7   276964.3
## [11,]  50     247613   320974.1   298042.5   279880.4
## [12,]  55     211351   243039.0   315045.0   292537.0
## [13,]  60     215140   205108.8   235861.0   305740.2
## [14,]  65     221764   204943.9   195388.1   224682.8
## [15,]  70     223506   204792.8   189259.9   180435.4
## [16,]  75     183654   194419.0   178141.1   164629.7
## [17,]  80     141990   142323.8   150666.2   138051.5
## [18,]  85     112424   131768.2   141959.9   151559.3
```

Example Questions:

a. What would the reference year be if the input were the mid-year population for 1993 (i.e., July 1st, 1993)?

The reference year would be 1993.5

b. How can we project the population for 100 years without using a loop?

We can use the exponential of the Leslie matrix to project the population for multiple periods. For example, to project the population for 100 years (i.e., 20 periods of 5 years each), we can calculate the exponential of the Leslie matrix raised to the power of 20 and then multiply it by the initial population vector:

```
projected          = (lESliE%^%20)%*%female[,"nNx_1993.0"]
```

c. Quantifying the projected number of births is no longer a necessary step when using a Leslie matrix. Since this quantity remains important for many research questions, how can we estimate the projected number of births by age of the mother when following the Leslie matrix approach?

The projected number of births by age of the mother can be estimated by multiplying the age-specific fertility rates by the exposure to the risk of childbearing. Keeping the same assumptions of the cohort component model, the exposure is calculated as the average number of the population at the beginning and at the end of the projection interval.

```
B                  = nFx*n/2*(female[,"nNx_1993.0"] + lESliE%*%female[,"nNx_1993.0"])
```

Which is the same as the following, but with a more compact code:

```
B                  = nFx*n/2*(diag(ages) + lESliE)%*%female[,"nNx_1993.0"]
```

Compulsory Questions:

a. Calculate the life expectancy at birth of this population.

```
sum(nLx)/radix
```

```
## [1] 81.05968
```

b. Calculate the Total Fertility Rate of this population.

```
sum(nFx*n)
```

```
## [1] 2.085
```

c. Calculate the Net Reproduction Rate of this population.

```
sum(nFx*nLx/radix)*(1/(1 + SRB))
```

```
## [1] 1.006424
```

d. Calculate the projected number of births from 1993.0 to 1998.0 by age group of the mother.

Short answer

```
current            = female[,"nNx_1993.0"]
B                  = n*nFx*(current + lESliE%*%current)/2
```

Same answer with a fancy table

```
femaleS            = female[,c("age",str_c("nNx_",sprintf("%.1f", year)),str_c("nNx_",sprintf("%.1f", yea
B                  = matrix(rowSums(femaleS[,c(2,3)])/2*n*nFx,ages,1)
```

```r
colnames(B)[1] <- str_c("B[",sprintf("%.1f", year),",",sprintf("%.1f", year + n),")")
femaleS           = cbind(femaleS,B)
```

e. Calculate the projected total number of births from 1993.0 to 1998.0.

```r
sum(B)
```

```
## [1] 604866.4
```

f. Calculate how many are projected to be male births.

```r
sum(B)*(SRB/(1 + SRB))
```

```
## [1] 309809.6
```

g. Calculate the projected number of births from 1993.0 to 1998.0 by cohort of the mother.

```r
B                 = matrix(rowSums(femaleS[,c(2,3)])*n/2*nFx,ages,1)
x                 = femaleS[,"age"]
cohort            = matrix(paste(as.character(year-x-n), as.character(year-x-1), sep = "-"),ages)
colnames(cohort)[1] <- "Birth Cohort"
B                 = matrix(femaleS[,str_c("nNx_",sprintf("%.1f", year))]*nGx,ages,1)
colnames(B)[1] <- str_c("Bc[",sprintf("%.1f", year),",",sprintf("%.1f", year + n),")")
femaleC           = data.frame(cohort,B)
print(femaleS)
```

```
##       age nNx_1993.0 nNx_1998.0 B[1993.0,1998.0)
##  [1,]   0     293395   293573.8           0.0000
##  [2,]   5     248369   293189.2           0.0000
##  [3,]  10     240012   248250.6           0.0000
##  [4,]  15     261346   239833.3       15035.3785
##  [5,]  20     285209   261014.9      123992.8321
##  [6,]  25     314388   284786.9      224540.7763
##  [7,]  30     281290   313781.7      167363.9042
##  [8,]  35     286923   280463.0       62554.3065
##  [9,]  40     304108   285576.2       10909.1576
## [10,]  45     324946   301730.7         470.0075
## [11,]  50     247613   320974.1           0.0000
## [12,]  55     211351   243039.0           0.0000
## [13,]  60     215140   205108.8           0.0000
## [14,]  65     221764   204943.9           0.0000
## [15,]  70     223506   204792.8           0.0000
## [16,]  75     183654   194419.0           0.0000
## [17,]  80     141990   142323.8           0.0000
## [18,]  85     112424   131768.2           0.0000
```

```r
print(femaleC)
```

```
##    Birth.Cohort Bc.1993.0.1998.0.
## 1     1988-1992            0.0000
## 2     1983-1987            0.0000
## 3     1978-1982         7194.9985
## 4     1973-1977        67090.7691
```

```
## 5     1968-1972       171466.3163
## 6     1963-1967       206067.9947
## 7     1958-1962       110033.8582
## 8     1953-1957        36916.4203
## 9     1948-1952         5852.2960
## 10    1943-1947          243.7095
## 11    1938-1942            0.0000
## 12    1933-1937            0.0000
## 13    1928-1932            0.0000
## 14    1923-1927            0.0000
## 15    1918-1922            0.0000
## 16    1913-1917            0.0000
## 17    1908-1912            0.0000
## 18    1903-1907            0.0000
```

Exercise 2: Cohort Component Model of Projections (two sexes, female dominant perspective, closed population)

In this exercise, we extend the one-sex model to project both sexes simultaneously. For now, we continue to assume a female-dominant perspective and a closed population. We use the same data, so there is no need to modify the setup or the definition of key parameters. The only change is to read the data for each sex separately and to construct an augmented Leslie matrix that accounts for both sexes.

The data:

```
female            = input[,c("age","nLx_f","nFx","nNx_f")]
male              = input[,c("age","nLx_m","nNx_m")]
```

Construction of the augmented Leslie matrix:

The Leslie matrix is consisting of four sub-matrices representing: (1) how the female population survives and is self-reproduced, i.e., to project female population using only female information, $lESliE_{ff}$; (2) how the male population survives, i.e., to project male population using only male information, $lESliE_{mm}$; (3) how the male population reproduces from the female population, i.e., to project the number of male births using female information, $lESliE_{fm}$; and (4) how the female population would reproduce from the male population, i.e., to project the number of female births using male information, $lESliE_{mf}$, which is not the case in a female dominant perspective.
Note that we already know how the female population survives and self-reproduces, so we can simply use the Leslie matrix from the Exercise 1. We then need to construct the remaining (two) sub-matrices, using as an initial condition that all entries are set to zero, as shown.

Initial conditions

```
lESliE_ff         = lESliE
lESliE_mm         = matrix(0,ages,ages)
lESliE_fm         = matrix(0,ages,ages)
lESliE_mf         = matrix(0,ages,ages)
```

Male population

The first step is to calculate the survivorship ratios using the ${}_nL_x$ column.

```
nLx               = male[,"nLx_m"]
nSx               = c(nLx[1]/(n*radix),nLx[2:(ages-1)]/nLx[1:(ages-2)],nLx[ages]/sum(nLx[(ages-1):ages]
```

The second step is to insert the multipliers needed to project the first age group of the male population,

based on the fertility schedule and the age distribution of women in the reproductive ages. Note that we have used the same ${}_nG_x$ values as in Exercise 1, but we now apply the male survivorship ratios ${}_nS_x$ and multiply by the sex ratio at birth (SRB).

```
lESliE_fm[1,]      = nGx*SRB/(1 + SRB)*nSx[1]
```

The third step is to insert the multipliers needed to project the remaining age groups, representing the survival of the male population. We simply use the same formulas as in Exercise 1, but applied to the male data.

```
diag(lESliE_mm[-1,]) = nSx[2:ages]
lESliE_mm[ages,ages] = nSx[ages]
```

Once the four sub-matrices are constructed (we didn't have to modify $lESliE_{mf}$), we can concatenate them to obtain the augmented Leslie matrix for projecting both sexes simultaneously, assuming female population on top. We also need to concatenate the population vectors so that we have a single input vector for the projection, as shown:

```
lESliE             = rbind(cbind(lESliE_ff,lESliE_mf),cbind(lESliE_fm,lESliE_mm))
current            = rbind(matrix(female[,"nNx_f"],ages),matrix(male[,"nNx_m"],ages))
```

Projecting the two sexes simultaneously:

We can now project the two sexes using one single operation, as we did in Exercise 1. The resulting (projected) population is a (column) vector of length $2 * ages$, where the first half corresponds to the projected female population.

```
projected          = lESliE%*%current
```

Let's produce a simple table of results, projecting the population 3 consecutive periods holding fertility and mortality constant (i.e., the Leslie matrix is always the same). We first retain only the relevant information (pop) and rename the current population using the reference year. The function $str_c()$ concatenates several strings, and we want the column names to display one digit of decimal precision using the function $sprintf($", as shown:

```
pop                = input[,c("age","nNx_m","nNx_f")]
colnames(pop)[colnames(pop) == "nNx_f"] <- str_c("nNx_f",sprintf("%.1f",year))
colnames(pop)[colnames(pop) == "nNx_m"] <- str_c("nNx_m",sprintf("%.1f",year))
```

We then use a loop to project the population iteratively, drawing on the most recent data at each step. The loop is defined over a sequence of years starting from the reference year and ending at the final projection year, with a step length equal to the width of the age intervals (i.e., $seq(year, year + 2 * n, n)$). At each step, we define the input vector "*current*" from the most recent data, and then compute the "*projected*" output vector by multiplying the Leslie matrix by the current population. We then extract the male and female projections, assign appropriate names, and concatenate the resulting tables to include the new set of projections, as shown:

```
for (t in seq(year, year + 2*n, n)) {
  current   = rbind(matrix(pop[,str_c("nNx_f",sprintf("%.1f",t))],ages),matrix(pop[,str_c("nNx_m",sprin
  projected = lESliE%*%current                                  #Projects the population.
  female    = matrix(projected[1:ages,],ages)                   #Extracts female population.
  colnames(female)[1] <- str_c("nNx_f",sprintf("%.1f",t + n)) #Gives a name to the projected female pop
  male      = matrix(projected[(1 + ages):(2*ages),],ages)      #Extracts male population.
  colnames(male)[1] <- str_c("nNx_m",sprintf("%.1f",t + n))   #Gives a name to the projected male popul
  pop       = cbind(pop,male,female)                            #Concatenates the table of results.
}
print(pop)
```

```
##       age nNx_m1993.0 nNx_f1993.0 nNx_m1998.0 nNx_f1998.0 nNx_m2003.0
##  [1,]   0      310189      293395   307798.32    293573.8   293693.20
##  [2,]   5      261963      248369   309903.63    293189.2   307515.15
##  [3,]  10      252046      240012   261800.43    248250.6   309711.31
##  [4,]  15      274711      261346   251600.84    239833.3   261338.04
##  [5,]  20      296679      285209   273793.84    261014.9   250760.84
##  [6,]  25      333726      314388   295485.57    284786.9   272692.47
##  [7,]  30      296774      281290   332266.09    313781.7   294192.95
##  [8,]  35      299391      286923   295052.76    280463.0   330339.00
##  [9,]  40      314295      304108   296870.88    285576.2   292569.16
## [10,]  45      338709      324946   310477.01    301730.7   293264.55
## [11,]  50      256066      247613   332337.58    320974.1   304636.65
## [12,]  55      208841      211351   248055.45    243039.0   321941.01
## [13,]  60      199996      215140   197597.62    205108.8   234700.88
## [14,]  65      197282      221764   182332.85    204943.9   180146.29
## [15,]  70      184234      223506   169672.10    204792.8   156815.11
## [16,]  75      133856      183654   143376.81    194419.0   132044.28
## [17,]  80       86732      141990    88211.05    142323.8    94485.26
## [18,]  85       49095      112424    58052.00    131768.2    62512.33
##       nNx_f2003.0 nNx_m2008.0 nNx_f2008.0
##  [1,]    280120.5   276991.27    264190.5
##  [2,]    293367.9   293423.01    279924.0
##  [3,]    293049.4   307324.31    293228.0
##  [4,]    248065.7   309164.31    292831.2
##  [5,]    239529.5   260465.53    247751.5
##  [6,]    260628.6   249752.13    239174.9
##  [7,]    284237.6   271499.56    260125.9
##  [8,]    312859.1   292486.68    283401.9
##  [9,]    279146.5   327558.38    311390.6
## [10,]    283343.7   289015.09    276964.3
## [11,]    298042.5   287747.98    279880.4
## [12,]    315045.0   295106.66    292537.0
## [13,]    235861.0   304608.66    305740.2
## [14,]    195388.1   213972.68    224682.8
## [15,]    189259.9   154934.56    180435.4
## [16,]    178141.1   122038.55    164629.7
## [17,]    150666.2    87017.12    138051.5
## [18,]    141959.9    67100.24    151559.3
```

Compulsory Questions:

a. Calculate the life expectancy at birth for each sex.

```r
sum(input[,"nLx_m"])/radix
```

```
## [1] 75.42094
```

```r
sum(input[,"nLx_f"])/radix
```

```
## [1] 81.05968
```

b. Calculate the resulting age-sex distribution of keeping constant the fertility and mortality regimes for a long period of time (i.e., stable-equivalent age distribution, ergodic solution).

```
current           = rbind(matrix(input[,"nNx_f"],ages),matrix(input[,"nNx_m"],ages))
current           = current/sum(current)
projected         = (lESliE%^%100)%*%current
projected         = projected/sum(projected)

female            = matrix(projected[1:ages,],ages)
colnames(female)[1] <- "nNx_f*"
male              = matrix(projected[(1 + ages):(2*ages),],ages)
colnames(male)[1] <- "nNx_m*"
ergodic           = cbind(input[,"age"],male,female)
print(ergodic)
```

```
##               nNx_m*     nNx_f*
##  [1,]  0 0.032824587 0.03130764
##  [2,]  5 0.032757943 0.03125091
##  [3,] 10 0.032701232 0.03120130
##  [4,] 15 0.032607202 0.03114342
##  [5,] 20 0.032462226 0.03106940
##  [6,] 25 0.032295714 0.03098894
##  [7,] 30 0.032118697 0.03089480
##  [8,] 35 0.031896922 0.03076973
##  [9,] 40 0.031593281 0.03059126
## [10,] 45 0.031174815 0.03031840
## [11,] 50 0.030554403 0.02991453
## [12,] 55 0.029565672 0.02932931
## [13,] 60 0.027942849 0.02843144
## [14,] 65 0.025446687 0.02705388
## [15,] 70 0.021861065 0.02495573
## [16,] 75 0.016994084 0.02168389
## [17,] 80 0.011186654 0.01678539
## [18,] 85 0.008333631 0.01799235
```

Exercise 3: Cohort Component Model of Projections (one sex, female dominant perspective, open population)

In this exercise, we extend the one-sex model to project a population that is open to migration. Since the underlying data remain unchanged, the setup and the definition of key parameters do not require any modification. We have already calculated the Leslie matrix for a female population that is self-reproducing, $lESliE_{ff}$ and we can keep this result representing a counterfactual scenario of a closed population. The only adjustment is to import the data containing the number of net migrants for the female population and to construct a modified Leslie matrix that explicitly incorporates migration.

The data:

```
lESliE_C          = lESliE_ff
female            = input[,c("age","nLx_f","nFx","nNx_f","nIx_f")]
```

Construction of the modified Leslie matrix to account for migration:

We calculate again the survivorship ratios using the ${}_nL_x$ column.

```
nLx                 = female[,"nLx_f"]
nSx                 = c(nLx[1]/(n*radix),nLx[2:(ages-1)]/nLx[1:(ages-2)],nLx[ages]/sum(nLx[(ages-1):ages]
```

Let's define a vector of age-specific net migration rates by dividing the number of net migrants by the population at time t. This is not to be consider an untenable assumption, given that half of all migrants are assumed to arrive (or depart) at the beginning of the projection interval. Thus, the total number of migrants could be inferred at the exact time t.

```
nKx                 = c(female[,"nIx_f"]/female[,"nNx_f"])
```

Migrants in reproductive ages will affect the projected number of births. The next step is to modify the function ${}_nG_x$ to incorporate specific assumptions about fertility and migration. We assume that migrants reproduce at the same rates as the domestic population, but we also need to account for the fact that migrants arrive at different times during the projection interval increasing (reducing) the risk of childbearing. The modified function ${}_nG_x$ is calculated as follows:

```
nGx                 = ((1 + nKx)*nFx + (1 + nKx/2)*c(nFx[2:ages]*nSx[2:ages],0))*n/2
```

We follow the same approach used in the previous models. We define the Leslie matrix as a square matrix whose order is equal to the number of age groups, and all entries are initially set to zero before the matrix is populated. We then insert the multipliers related to survivorship and reproduction, taking into account that migration interacts with both components. Let's begin with the first row, which corresponds to the reproductive contribution of this open population, as shown:

```
lESliE              = matrix(0,ages,ages)
lESliE[1,]          = nGx*1/(1 + SRB)*nSx[1]
```

As in the previous models, we populate the first sub-diagonal using the survivorship ratios and, now, the effect of migration. All populations decrease through mortality, but open populations may increase (or decrease less) when net migration is positive. Conversely, when the number of out-migrants exceeds the number of in-migrants, the force of attrition is intensified by the effect of negative net migration.

```
diag(lESliE[-1,]) = (1 + nKx[1:ages-1]/2)*nSx[2:ages]
```

We then populate the main diagonal of the Leslie matrix, which now accounts for the effect of migrants who arrive at the end of the projection period.

```
diag(lESliE[])      = diag(lESliE[]) +  nKx/2
```

Finally, we adjust the survivorship ratio for the open-ended age interval in the last entry of the last column, which has always required special attention. In this case, the adjustment accounts for the effect of net migration.

```
lESliE[ages,ages] = lESliE[ages,ages] + (1 + nKx[ages]/2)*nSx[ages]
```

Projecting a female population open to migration:

As in the previous models, projecting a population is reduced to a single operation: multiplying a column population vector by the corresponding Leslie matrix, as shown:

```
current             = female[,"nNx_f"]
projected           = lESliE%*%current
```

Let's produce a simple table of results, projecting the population for three consecutive periods while holding fertility and mortality constant (i.e., the Leslie matrix remains unchanged). To quantify the effect of migration on population dynamics, let's project the population twice—first assuming the population is open, and then

assuming it is closed to migration.
We can create two structures $female_O$ and $female_C$, one for each scenario. They are initially identical and retain only the essential information.

```
female_O          = input[,c("age","nNx_f")]
colnames(female_O)[colnames(female_O) == "nNx_f"] <- str_c("nNx_",sprintf("%.1f", year))
female_C          = female_O
```

We then use a loop to project the population iteratively, as we have done in previous models, using as an input the most recent data at each step. The loop is defined over a sequence of years (i.e., $seq(year, year + 2 * n, n)$). At each step, we define the input vector "*current*" from the most recent data, and then compute the "*projected*" output vector by multiplying the Leslie matrix by the current population. We then extract the male and female projections, assign appropriate names, and concatenate the resulting tables to include the new set of projections, as shown:

```
for (t in seq(year, year + 2*n, n)) {
  projected = lESliE%*%female_O[,str_c("nNx_",sprintf("%.1f", t))]   #Projects the open population.
  colnames(projected)[1] <- str_c("nNx_",sprintf("%.1f", t + n))     #Gives a name to the projected dat
  female_O  = cbind(female_O,projected)                              #Concatenates the table of results

  projected = lESliE_C%*%female_C[,str_c("nNx_",sprintf("%.1f", t))] #Projects the open population.
  colnames(projected)[1] <- str_c("nNx_",sprintf("%.1f", t + n))     #Gives a name to the projected dat
  female_C  = cbind(female_C,projected)                              #Concatenates the table of results
  }

print(female_O)
```

```
##       age nNx_1993.0 nNx_1998.0 nNx_2003.0 nNx_2008.0
##  [1,]   0     293395   302391.8   295972.5   285684.4
##  [2,]   5     248369   298681.8   308197.4   301787.3
##  [3,]  10     240012   252007.1   302800.1   312746.7
##  [4,]  15     261346   244149.5   256046.3   307277.3
##  [5,]  20     285209   268266.6   250644.3   262360.4
##  [6,]  25     314388   293515.0   276050.4   257940.6
##  [7,]  30     281290   320623.7   299900.8   282040.2
##  [8,]  35     286923   284767.5   324355.0   303709.7
##  [9,]  40     304108   288031.3   285827.3   325439.2
## [10,]  45     324946   303166.3   287131.4   284910.8
## [11,]  50     247613   322062.3   300673.5   284761.0
## [12,]  55     211351   243989.1   317287.0   296394.2
## [13,]  60     215140   205841.4   237564.9   308887.7
## [14,]  65     221764   205516.1   196625.4   226880.1
## [15,]  70     223506   205270.0   190228.6   181992.9
## [16,]  75     183654   194771.2   178901.1   165790.6
## [17,]  80     141990   142565.0   151187.9   138886.8
## [18,]  85     112424   131965.7   142408.1   152298.8
```

```
print(female_C)
```

```
##       age nNx_1993.0 nNx_1998.0 nNx_2003.0 nNx_2008.0
##  [1,]   0     293395   293573.8   280120.5   264190.5
##  [2,]   5     248369   293189.2   293367.9   279924.0
##  [3,]  10     240012   248250.6   293049.4   293228.0
##  [4,]  15     261346   239833.3   248065.7   292831.2
##  [5,]  20     285209   261014.9   239529.5   247751.5
##  [6,]  25     314388   284786.9   260628.6   239174.9
```

```
##  [7,]  30     281290   313781.7   284237.6   260125.9
##  [8,]  35     286923   280463.0   312859.1   283401.9
##  [9,]  40     304108   285576.2   279146.5   311390.6
## [10,]  45     324946   301730.7   283343.7   276964.3
## [11,]  50     247613   320974.1   298042.5   279880.4
## [12,]  55     211351   243039.0   315045.0   292537.0
## [13,]  60     215140   205108.8   235861.0   305740.2
## [14,]  65     221764   204943.9   195388.1   224682.8
## [15,]  70     223506   204792.8   189259.9   180435.4
## [16,]  75     183654   194419.0   178141.1   164629.7
## [17,]  80     141990   142323.8   150666.2   138051.5
## [18,]  85     112424   131768.2   141959.9   151559.3
```

Compulsory Questions:

a. Can we expect life expectancy to change because of migration?

b. Can we expect the TFR to change because of migration?

c. Can we expect the NRR to change because of migration?

d. Calculate the projected number of births from 1993.0 to 1998.0 by age group of the mother and discriminating by migrants and nationals.

e. What change can we expect in the projected number of births if net migration is negative (i.e., ${}_nI_x < 0$)?.

f. Which line of this code would you modify if migrants brought their own age-specific fertility rates?

g. Which line of this code would you modify if migrants imported their own mortality schedule (i.e., ${}_nL_x$)?